

# Isaac McDowell

New Orleans, LA • isaac@isaacmcdowell.dev • isaacmcdowell.dev • linkedin.com/in/isaac-mcdowell

## Education

**Florida State University** – BS in Computer Science

May 2026

- **Relevant Coursework:** *Data Structures & Algorithms, Software Engineering, Database Systems, Object-Oriented Programming, Introduction to AI / Machine Learning, Web Development*

## Projects

**Compass - Personal Dashboard** (Active Development) | *React, Node.js, Express, MongoDB*

[Live Demo](#) · [GitHub](#)

Solo – full-stack development

- Built a full-stack habit and task tracker for managing daily routines, recurring tasks, and progress over time.
- Designed a two-collection MongoDB schema separating habits from completion logs, enforcing one completion per habit per day.
- Built task completion with optimistic UI updates, a short undo window, and automatic rollback on failed requests.
- Calculated streaks at page load using each user's timezone instead of a scheduled reset job, keeping reads fast and avoiding extra background processing.

**Smart Fitness App** | *Flask, PostgreSQL, Python, Chart.js*

[Live Demo](#) · [GitHub](#)

Solo – full-stack development

- Built and deployed a Flask/PostgreSQL fitness tracker for meal logging, workouts, and weight monitoring.
- Built a rule-based insight engine that analyzes nutrition, workout, and weight trends to generate personalized coaching feedback.
- Integrated the USDA FoodData API for real-time food search with automatic macro extraction for 600,000+ items.
- Built Chart.js dashboards visualizing macronutrient breakdown and weight trends over time.

**Tokyo Bloom - Restaurant Web App** | *HTML/CSS, JavaScript, PHP, MySQL*

[GitHub](#)

Team of 3 – full-stack development

- Developed responsive frontend UI, including animated landing components and interactive Google Maps integration.
- Implemented reservation validation logic to prevent double bookings and improve scheduling consistency.
- Added security protections against CSRF, XSS, and SQL injection.
- Led an MVC refactor that improved the team's code organization and maintainability.

## Experience & Involvement

**Association for Computing Machinery (ACM)**, Member

2022 – 2025

- Participated in competitive programming competitions and workshops, practicing algorithm design and problem-solving under time constraints.

**Cybersecurity Club at FSU**, Member

2022 – 2024

- Competed in CTF events and attended workshops on web security, cryptography, and penetration testing.

**Front Desk Staff**, YMCA – Yulee, FL

2021

- Managed front desk operations, including member check-ins, phone and in-person inquiries, and coordination with staff in a fast-paced environment.

## Skills

**Languages:** JavaScript, Python, C++, HTML/CSS, SQL

**Frameworks:** React, Node.js, Express, Flask, Bootstrap, Tailwind CSS, Chart.js

**Tools:** MongoDB, MySQL, SQLite, Linux, Git, Docker, Vercel, Railway